

# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer

By: Sebastion Mauldin - [Sebastion.mauldin@gmail.com](mailto:Sebastion.mauldin@gmail.com)  
5/28/2025

## Table of Contents

- Introduction
- Solution Architecture Design
- Virtual Machine Configuration
- Virtual LAN Configuration
- Cloud Integration
- Challenges and Mitigations
- Conclusion and Recommendation

## Introduction

This project simulates the deployment of a secure network infrastructure for a growing small business. Acting as the Network Administrator, I designed and implemented a multi-subnet enterprise network using Cisco Packet Tracer, configured routing between departments, deployed core network services, and validated end-to-end connectivity.

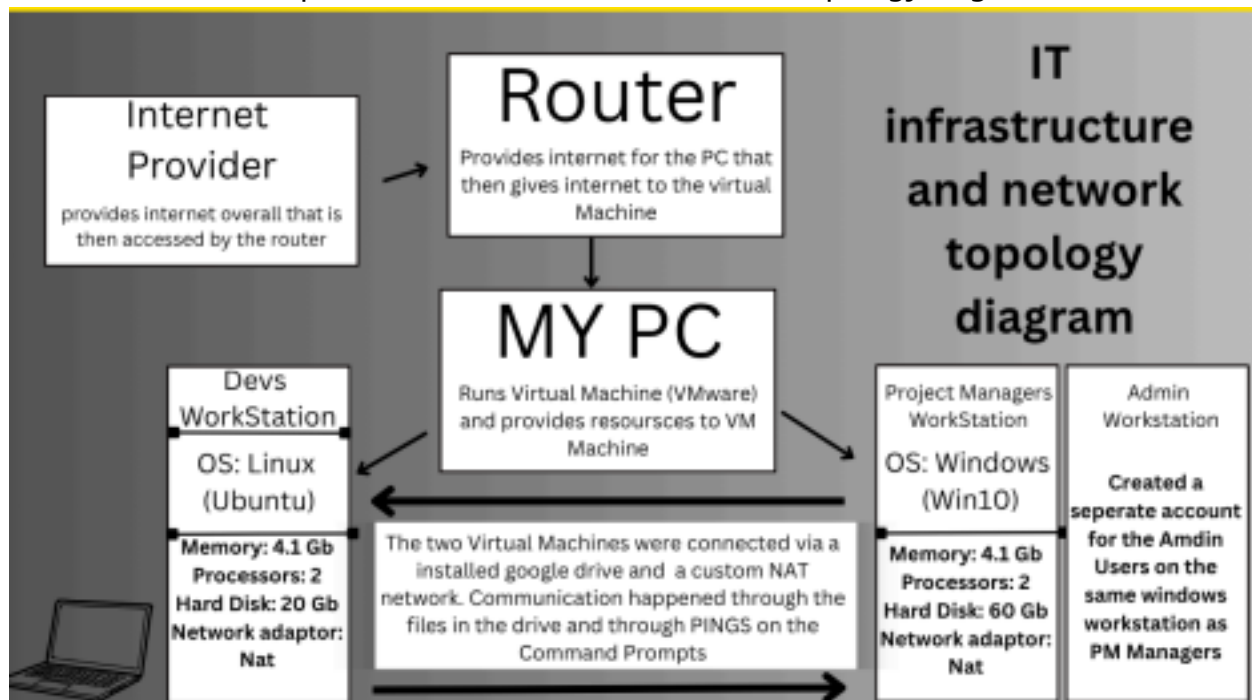
The objective was to demonstrate foundational networking concepts used in enterprise IT and cybersecurity environments including:

- IP Addressing
- Subnetting
- Dynamic Routing (RIPv2)
- DHCP
- DNS
- Network Verification
- Troubleshooting
- Documentation

# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer

## Solution Architecture Design

- Present a complete IT infrastructure and network topology diagram.



## Virtual Machine Configurations

Describe the configurations for each virtual machine:

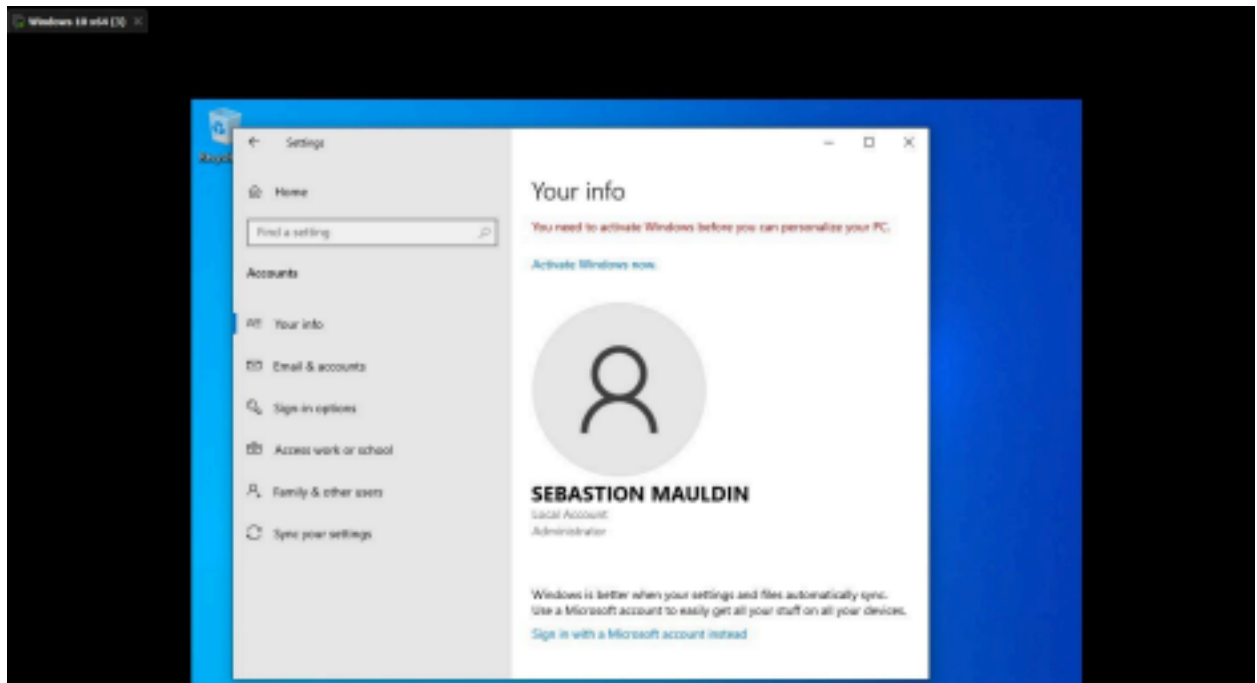
**Windows Workstation** (This would be an example for administrative and project management roles): First I downloaded and installed windows to the VMware Workstation. I had trouble for a bit with the install due to the OS insisting that I input a Licence. I was able to bypass this step, by hitting Install OS Later in the initial VWare install screen that appears when you try to initially set up a new machine. From then I simply went to the settings and directed the VMware to the Windows intall ISO and it worked. I hit simple setup and I went through the installation process. After it was completed created an administrative account for the admins and a Project

# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer

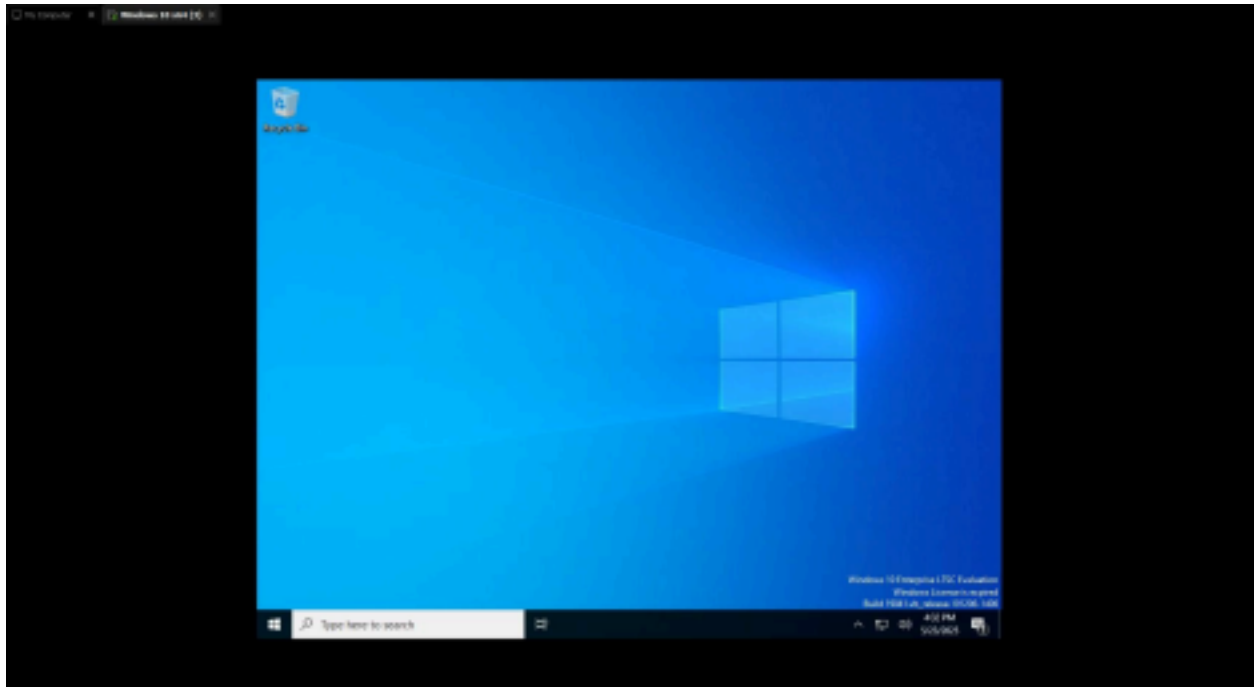
management account for the Project Managers.

## Specs:

- Memory 4.1gb
- Processors 2
- Hard Disk: 60 Gbs
- Network adaptor: Nat



# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer

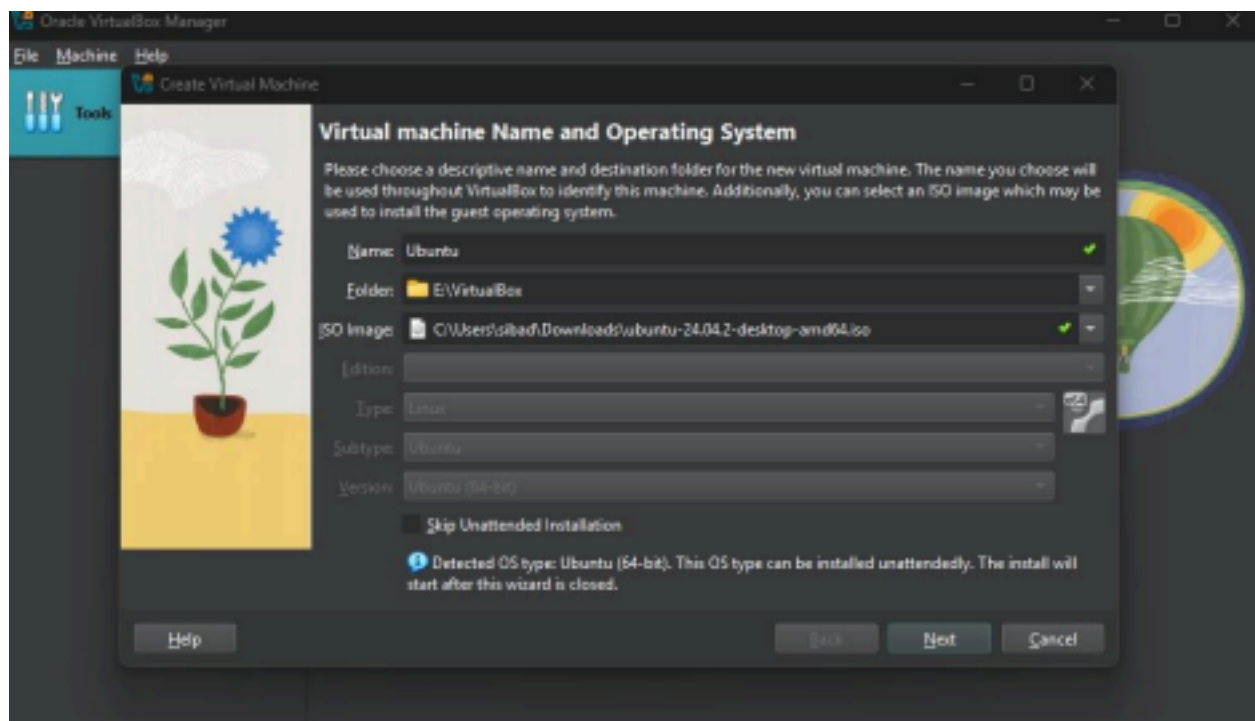


**Linux Workstation** (This would be an example for developer roles): for the Linux workstation I downloaded Ubuntu and used the downloaded disc image file to install the Ubuntu version of Linux. Unlike with the Windows install this one went a lot smoother. I was able to install and create an account for the Dev team.

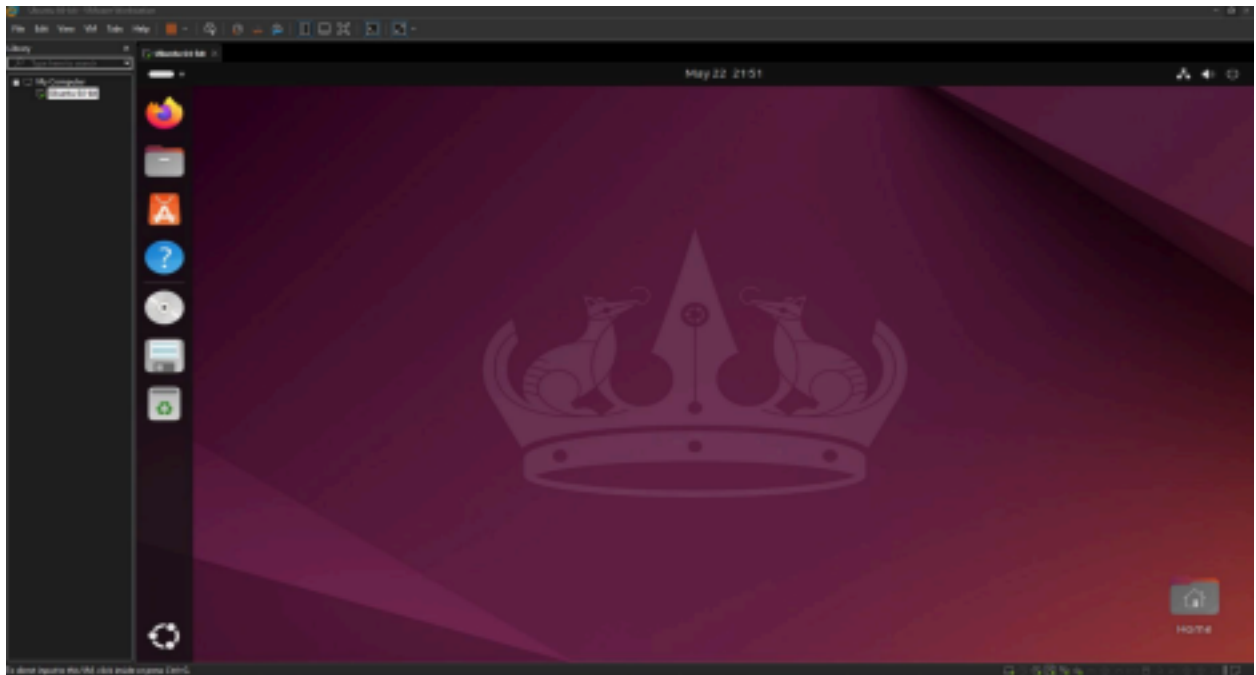
## Specs:

- Memory 4.1gb
- Processors 2
- Hard Disk: 60 Gbs

# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer



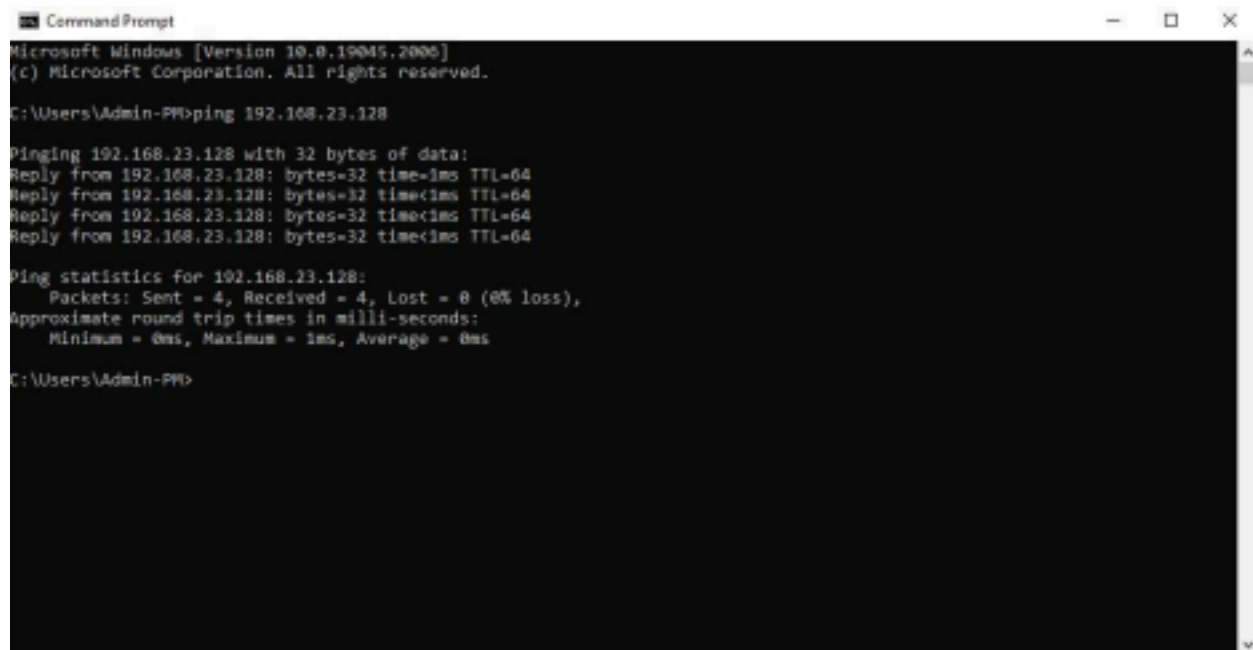
# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer



## Virtual LAN Configuration

- Explain how the private virtual LAN was set up for secure internal communication. In order to accomplish this I went into the VMware Workstation and change the settings to add a new network. I chose an Unused one and then configured it to host-only (but not to allow host access). After shutting down each OS I went to the network settings for each and chose the new custom virtual network. Afterwards I went in the terminal for Ubuntu and the Command Prompt for Windows and I set the new static IPs for each. Then I used the new IP addresses to Ping each account on their respective operating systems. It didn't work at first, but after editing the firewall settings in Windows (Enabling File and Printer Sharing in Windows Defender Firewall) I was able to have a successful PING!!!! Booyah!

# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer



```
Command Prompt
Microsoft Windows [Version 10.0.19045.2000]
(c) Microsoft Corporation. All rights reserved.

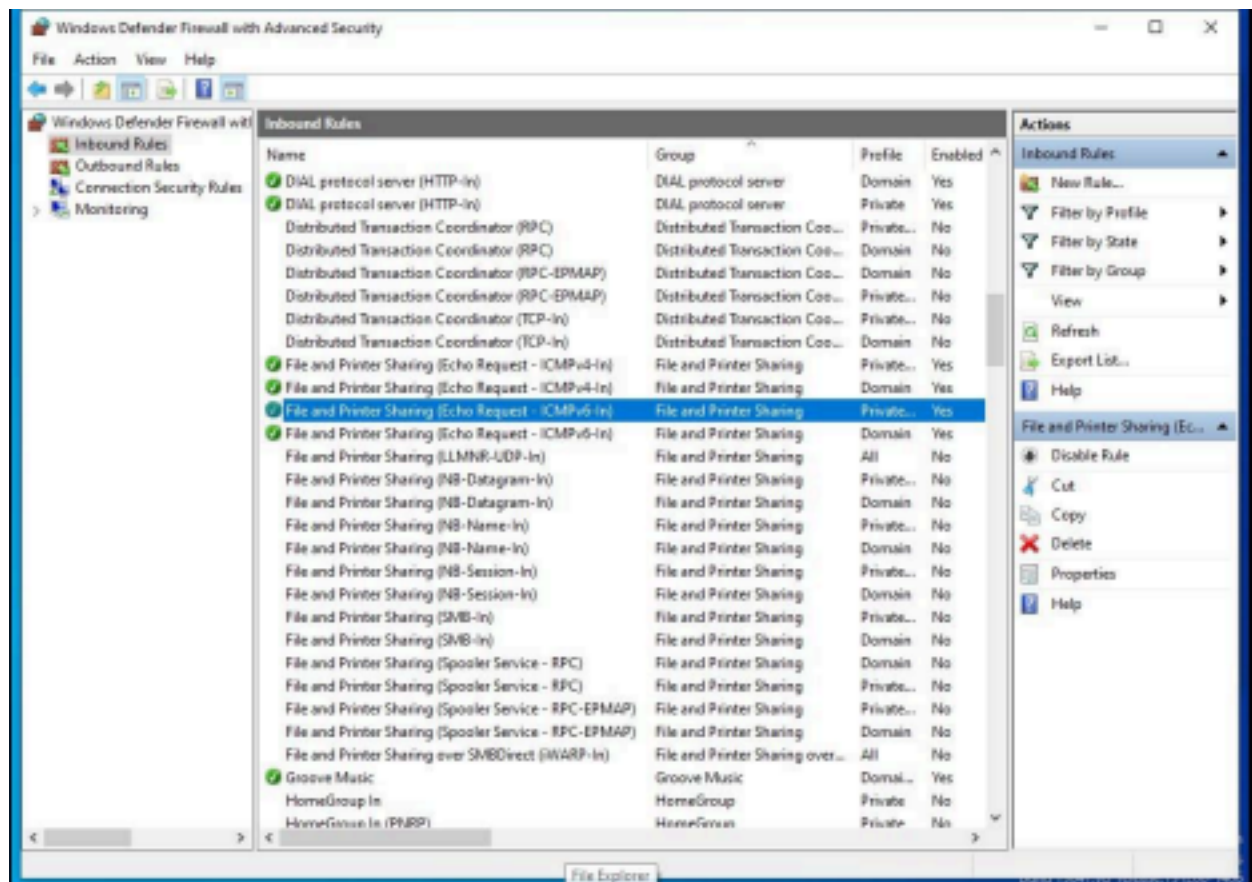
C:\Users\Admin-PR>ping 192.168.23.128

Pinging 192.168.23.128 with 32 bytes of data:
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64
Reply from 192.168.23.128: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.23.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Admin-PR>
```

# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer

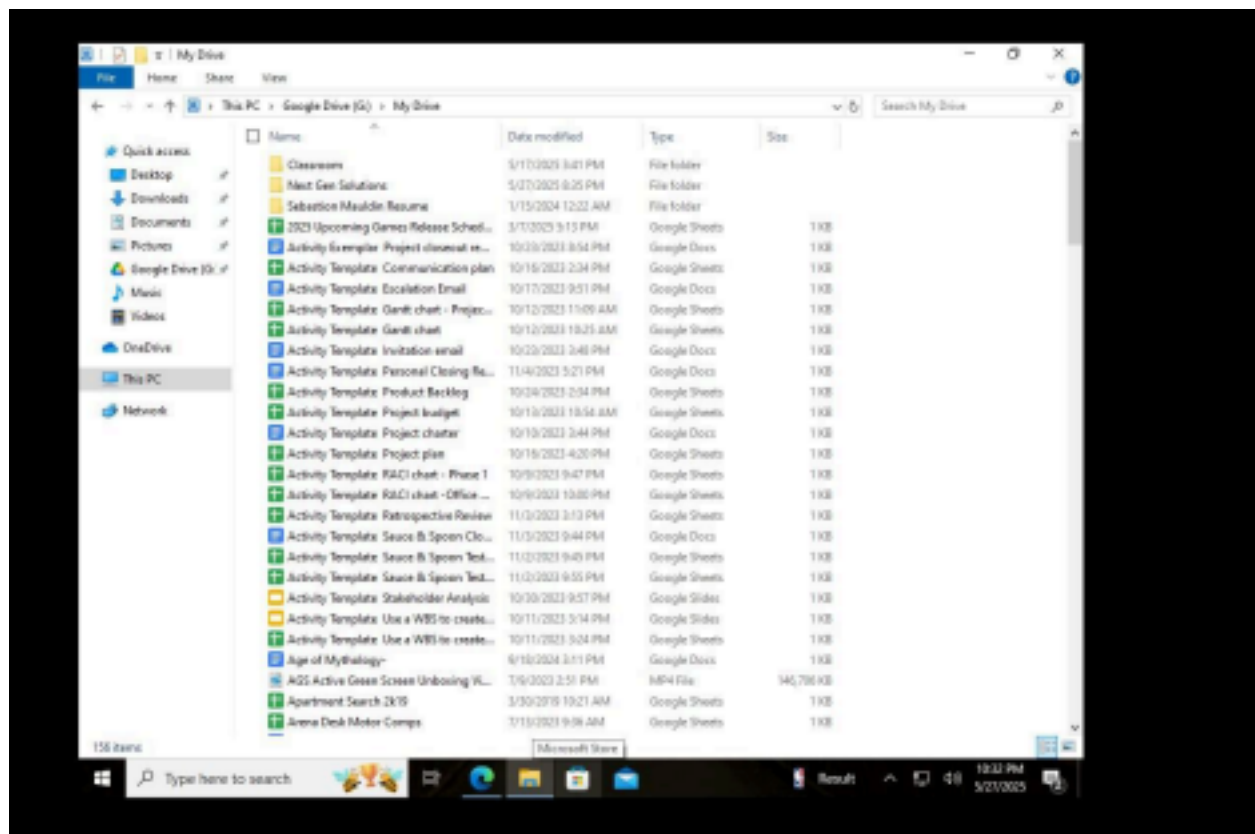


## Cloud Integration

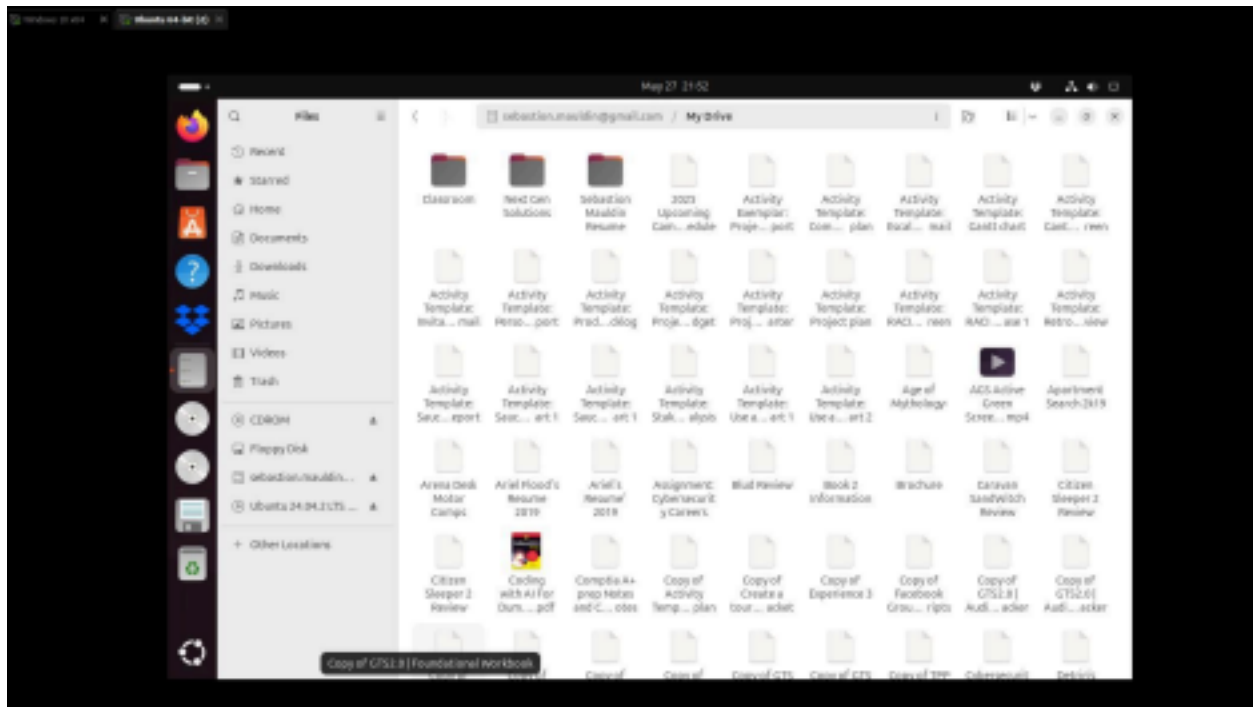
For cloud integration, I went with Google Drive. It is easy to access on each virtual Machine. For windows there is a google drive app that you can download online. From there it is simply installing the program on the computer and then logging in. On Linux, I discover that there wasn't a native Google drive app. So I simply logged in to my Google account in the settings and that gave me access to the google drive. Overall Google Drive was accessed, and each account was able to look in a folder that was created to show cloud access and access the google doc file and image that was in the folder.



# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer



# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer



## Challenges and Mitigations

- Describe any technical or design challenges you encountered. I experience a challenge in every single level of this assignment. I wasn't able to get Oracle Virtualbox working (by working I mean I received an error message when trying to install any OS). I had to troubleshoot how to get passed to Windows License requirements during its install. I messed up the NAT settings for VMware so I had to google how to fix that. I didn't know what a success ping looked like, so I had to figure that out as well as find out how to enable it with the Windows defender still on, because I did want that layer of safety still on the Windows Virtual machine.
- I went through all of these challenges and had to google and youtube my way through all of them. It was tedious and it felt like an uphill battle, but I got it done.

# Enterprise Network Infrastructure Deployment Using Cisco Packet Tracer

## Conclusion and Recommendations

- Summarize the effectiveness of your infrastructure design. I feel as though the infrastructure is good, but could be better. I was able to provide the office personal with a way to work in their respective operating systems as well as provide communication between the devices via a network and cloud integration. As for how I can improve this system, I think that as I learn more about security I could beef up the Windows side and the Ubuntu side to further ensure that there is less of a chance of an external attack.